



Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

The Big 4: Deriving the Kinematic Equations

Two definitions go in. Four equations come out.

Where the Big 4 come from

Every equation on this sheet is built from **two definitions** you already have.

Average velocity average velocity = $\frac{s}{t}$

and, when the acceleration is uniform, average velocity = $\frac{u + v}{2}$

Acceleration $a = \frac{\Delta v}{t}$ where $\Delta v = v - u$

s	u	v	a	t
distance	initial velocity	final velocity	acceleration	time
m	m/s	m/s	m/s ²	s

Reading the page. The physics runs down the left. The right-hand column does the **same algebra with plain letters**, and the same colour marks the same quantity. An arrow shows a piece of algebra moving into the next line. Each new equation ends in a **red box** with its number, and the **black box** names the one quantity it leaves out.

Equation 1

average velocity = $\frac{s}{t}$

average velocity = $\frac{u + v}{2}$

same algebra, plain letters

$a = b$ $a = c$

$$\frac{s}{t} = \frac{u + v}{2}$$

$b = c$

solve for s

$$\frac{s}{t} = \frac{1}{2}(u + v)$$

①

$$s = \frac{1}{2}(u + v)t$$

no a

Equation 2

$$a = \frac{\Delta v}{t}$$

$$\Delta v = \boxed{a t}$$

$$\Delta v = v - u$$

$$\Delta v = \boxed{v - u}$$

$$a t = v - u$$

same algebra, plain letters

$$a = \frac{b}{c} \quad b = d - e$$

$$b = a c \quad b = d - e$$

$$a c = d - e$$

solve for v

$$\textcircled{2} \quad \boxed{v = u + a t}$$

$$\boxed{\text{no } s}$$

Equation 3

— $\textcircled{1} + \textcircled{2}$, solve for s

$$\textcircled{2} \quad v = u + a t$$

$$\textcircled{1} \quad s = \frac{1}{2} (u + v) t$$

$$s = \frac{1}{2} (u + u + a t) t$$

solve for s

$$s = \frac{1}{2} (2 u + a t) t$$

$$s = \frac{1}{2} (2 u t + a t^2)$$

$$\textcircled{3} \quad \boxed{s = u t + \frac{1}{2} a t^2}$$

$$\boxed{\text{no } v}$$

same algebra, plain letters

$$a = b$$

$$c = \frac{1}{2} (d + a) e$$

$$c = \frac{1}{2} (d + b) e$$

Equation 4

— ① + ②, solve for v

same algebra, plain letters

$$\textcircled{1} \quad s = \frac{1}{2} (u + v) t$$

isolate t

$$(2s) = (u + v) (t)$$

$$t = \frac{2s}{u + v}$$

$$\textcircled{2} \quad v = u + at$$

$$v = u + a \left(\frac{2s}{u + v} \right)$$

$$v = u + \frac{2as}{u + v}$$

$$(v - u) = \frac{2as}{(u + v)}$$

$$(u + v)(v - u) = 2as$$

multiply out the brackets

$$v^2 - u^2 = 2as \text{ the form on your AQA equation sheet}$$

$$\textcircled{4} \quad v^2 = u^2 + 2as$$

$$a = bc$$

$$c = \frac{a}{b}$$

$$d = e + fc$$

$$d = e + f \left(\frac{a}{b} \right)$$

$$d = e + \frac{fa}{b}$$

$$(d - e) = \frac{fa}{b}$$

$$b(d - e) = fa$$

$$\begin{aligned} (u + v)(v - u) \\ = \cancel{uv} - u^2 + v^2 - \cancel{uv} \\ = v^2 - u^2 \end{aligned}$$

no t

The Big 4

— pick the one with **no** ...

	Equation	Use it when there is ...
①	$s = \frac{1}{2} (u + v) t$	no a
②	$v = u + at$	no s
③	$s = ut + \frac{1}{2} at^2$	no v
④	$v^2 = u^2 + 2as$	no t

Choosing one. Write down what you are given and what you want. The one quantity left over, neither given nor asked for, is in the right-hand column: that row is your equation. ④ is on your AQA equation sheet as $v^2 - u^2 = 2as$; ①, ② and ③ come from the same two definitions, so they are yours to use too.